

Sprint 1 Retrospective Report

Software Engineering - SVNC.00.308

Project: StayEase Hotel Reservation System
Team: Cat Bacillus (Kyrylo Pryiomyshev, Danylo Chernov, Daniil Ustuhov)
Date: March 2, 2026
Version: 1.0

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Sprint Number: 1

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Facilitator: Kyrylo Pryiomyshev

What Went Well

1. Clear task breakdown enabled parallel work

Example: We broke US-01 (Search Rooms) into three tasks: Kyrylo built the API endpoint, Danylo built the search UI, and Daniil prepared test cases. All three tasks progressed simultaneously, and US-01 was fully done in 25 minutes.

Why it worked: Small, independent tasks allowed all team members to contribute in parallel without blocking each other.

2. PostgreSQL date handling simplified availability logic

Example: Using PostgreSQL's OVERLAPS operator for date range queries, Kyrylo implemented the room availability check in a single SQL query instead of writing complex application-level logic. This saved approximately 20 minutes.

Why it worked: Choosing the right technology for the problem domain paid off. Our Design Documentation identified PostgreSQL's date handling as a key advantage, and Sprint 1 validated that decision.

3. End-to-end testing caught a critical bug early

Example: When Daniil tested the full booking flow (search → view → login → book), he discovered that the reservation API returned per-night rate instead of total price. Kyrylo fixed it in 5 minutes because the code was fresh.

Why it worked: Testing the complete workflow immediately after implementation meant the bug was caught while context was still fresh.

What Can We Improve

1. Overcommitted on sprint capacity

Example: We committed to 21 points (5 stories) but completed only 18 points (4 stories). US-05 was left incomplete because concurrent transaction handling was more complex than the S (3 point) estimate suggested.

Impact: We ended the sprint with an incomplete story, adding uncertainty to Sprint 2 planning and making our demo feel incomplete.

2. Underestimated technical complexity of US-05

Example: US-05 was estimated as Small (3 points) assuming a simple SQL OVERLAPS check would suffice. In reality, preventing double bookings under concurrent access requires transaction-level isolation (SELECT FOR UPDATE).

Impact: The unfinished story blocked a key data integrity requirement.

3. Testing was front-loaded on Daniil

Example: All testing tasks were assigned to Daniil, creating a bottleneck. While Kyrylo and Danylo were implementing US-04, Daniil had to wait for their code before testing.

Impact: Uneven workload distribution meant one team member was underutilized during implementation peaks.

Action Items for Sprint 2

Action Item 1

Action: Reduce Sprint 2 commitment to 16-18 points based on actual Sprint 1 velocity (18 points completed)

How: Use Sprint 1 velocity as the baseline. Only commit to stories we are confident we can complete.

Success metric: We complete 100% of committed stories in Sprint 2 without rushing.

Action Item 2

Action: Distribute testing across all team members instead of assigning only to Daniil

How: Each developer tests their own feature immediately after implementation. Daniil focuses on integration testing.

Success metric: No team member has idle time waiting for others.

Action Item 3

Action: Flag high-risk stories during planning and add a complexity buffer

How: Discuss each story's technical risks explicitly. If concurrency or transactions are involved, bump the estimate by one size.

Success metric: No stories left incomplete due to underestimated complexity.

Lessons Learned

The 90-minute timebox forced us to make pragmatic decisions rather than over-engineering solutions. When we realized US-05 needed transaction-level locking, we deliberately deferred it rather than rushing a half-baked implementation. Delivering four fully working features was better than five partially working ones. This is a key Agile insight: working software over comprehensive scope.

We also learned that velocity is discovered empirically, not guessed. Our initial estimate of 21 points was optimistic by about 15%. In Sprint 2, we will plan based on our actual demonstrated capacity of 18 points, which should lead to a more realistic and less stressful sprint. The contrast with Waterfall is striking: in Part II, we planned the entire project upfront with estimates we had no way to validate.

Team Sign-Off

Kyrylo Pryiomyshev: _____ Date: March 2, 2026

Danylo Chernov: _____ Date: March 2, 2026

Daniil Ustuhov: _____ Date: March 2, 2026